

L^AT_EX テンプレート

電気通信大学 IX 類
9999999 無敵太郎

2112 年 9 月 3 日作成

0.1 図の挿入

図は`\includegraphics` を用いると挿入できる．大きすぎるなら縮尺を変更して収めることも可能だ．図 1 に見本を示しておこう．

0.2 プログラムのソースコード

コード 1: サンプルプログラム

```
1 import re
2 from decimal import Decimal
3 from fractions import Fraction
4
5 class Input_matrix:
6     def __init__(self):
7         self.A = [[0]]
8         self.B = [[0]]
9         self.C = [[0]]
10        self.n = 0
11
12    def tableinput(self,a):
13        name = a
14        row = 0
15        row_count = 0
16        i = 0
17        j = 0
18        pre = []
19        print(f"行列 {name} を入力\n入力例:1 1.2 2/3 351\n endと入力して行入力終了")
20
21        while row != "end":
```

```

22         row = input(str(i+1)+ "行目:")
23         if row == "end":
24             break
25         pre.insert(i, row)
26         row_count = len(pre[0].split())
27
28         if re.search(r"^\d./ -]", row) and row != "end":
29             print("数字と-./以外入れるなポケ")
30             del pre[i]
31             continue
32
33         elif row_count != len(pre[int(i)].split()):
34             print("列数がちげーよポケ")
35             del pre[i]
36             continue
37
38         print(*pre, sep="\n")
39         i = i + 1
40
41     a = [[Fraction(x) for x in s.split()] for s in pre]
42     print(f"\n行列{name}の入力終了")
43     print(*a, sep="\n")
44     return a
45
46     def numberinput(self,n):
47         while True:
48             n = input("倍数n:")
49             if re.search(r"^\d./ -]", n):
50                 print("数字と-./以外入れるなポケ")
51                 continue
52             break
53         return n
54
55
56     class Tablecalculate:
57         def __init__(self):
58             self.A = [[0]]
59             self.B = [[0]]
60             self.C = [[0]]
61             self.n = 0
62
63         def simplify(self,a):

```

```

64     i = 0
65     j = 0
66     k = 0
67     process_row = 0
68     stop = False
69
70     gyoretsushiki = False
71     m = 1
72
73     while process_row < len(a):
74         i = process_row
75         j = 0
76         #Step1
77         while a[i][j] == 0:
78             if i == len(a)-1 and a[i][j] == 0:
79                 i = process_row
80                 j = j + 1
81                 if j > len(a[0]) - 1:
82                     stop = True
83                     break
84                 continue
85             if stop == True:
86                 break
87             i = i + 1
88             if j > len(a[0])-1:
89                 break
90             print(f"a{i+1,j+1}に注目")
91             tmp_i = i
92             tmp_j = j
93         #Step2
94         k = process_row
95         while a[k][tmp_j] == 0:
96             k = k + 1
97         if process_row+1 != tmp_i+1:
98             for j in range(len(a[0])):
99                 tmp = a[process_row][j]
100                 a[process_row][j] = a[tmp_i][j]
101                 a[tmp_i][j] = tmp
102             print(f"{process_row+1}行目 ⇔ {tmp_i+1}行目")
103             if gyoretsushiki == True:
104                 m = -m
105             print(f"倍数m:{m}")

```

```

106         tmp_i = process_row
107         self.output(a)
108     #Step3
109     if a[tmp_i][tmp_j] != 1:
110         k = a[tmp_i][tmp_j]
111         print(f"{tmp_i + 1}行目 / {k}")
112         if gyoretsushiki == True:
113             m = m * k
114             print(f"倍数m:{m}")
115         for j in range(len(a[0])):
116             a[tmp_i][j] = Fraction(a[tmp_i][j], k)
117
118         self.output(a)
119     #Step4
120     for i in range(process_row if gyoretsushiki == True else 0, len(a)):
121         if a[i][tmp_j] != 0 and i != tmp_i:
122             k = a[i][tmp_j]
123             for j in range(len(a[0])):
124                 a[i][j] = a[i][j] - a[tmp_i][j] * k
125             print(f"{i + 1}行目 - {tmp_i + 1}行目 × {k}")
126
127         self.output(a)
128         process_row = process_row + 1
129         print("row=", process_row)
130         if gyoretsushiki == True:
131             print(f"倍数m:{m}")
132     return a
133
134     def output(self, a):
135         b = [[0 for j in range(len(a[0]))] for i in range(len(a))]
136         for j in range(len(a[0])):
137             for i in range(len(a)):
138                 b[i][j] = str(a[i][j])
139         print(" ", *b, sep="\n")
140
141
142 InputMatrix = Input_matrix()
143
144 A = InputMatrix.tableinput("A")
145 TableCalculate = Tablecalculate()
146 result = TableCalculate.simplify(A)
147 print("計算結果:")

```

```
148 print(*result, sep="\n")
```
